ARTICLE IN PRESS

JACC: HEART FAILURE VOL. ■, NO. ■, 2026

PUBLISHED BY ELSEVIER

ORIGINAL RESEARCH

Cost-Effectiveness of Tirzepatide for the Management of Heart Failure With Preserved Ejection Fraction and Obesity

Tadesse M. Abegaz, PhD,[a] Gabriel Frietze, PhD,[a] Akshaya Srikanth Bhagavathula, PhD,[b] Askal Ayalew Ali, PhD[c]

ABSTRACT

BACKGROUND More than 80% of patients with heart failure with preserved ejection fraction (HFpEF) are overweight or obese. However, cost-effective treatments that target both conditions remain limited.

OBJECTIVES The goal of this study was to evaluate the cost-effectiveness of tirzepatide compared with standard care alone for the management of HFpEF in patients with obesity in the United States.

METHODS A Markov state-transition model was developed to project clinical and economic outcomes over a 20-year time horizon using data from the SUMMIT (A Study of Tirzepatide [LY3298176] in Participants With Heart Failure With Preserved Ejection Fraction [HFpEF] and Obesity) trial. The model included 3 health states: stable HFpEF, worsening heart failure, and death. Both costs and health outcomes were discounted at an annual rate of 3%. The primary outcome was the incremental cost-effectiveness ratio. Deterministic sensitivity analysis was performed to assess key drivers of cost-effectiveness, and probabilistic sensitivity analysis was conducted using 10,000 Monte Carlo simulations to assess uncertainty in model parameters. Outcomes were expressed as total quality-adjusted life years (QALYs), with the unit of analysis being per individual.

RESULTS Tirzepatide treatment yielded 7.27 (95% uncertainty interval [UI]: 3.20-13.70) QALYs compared with 6.63 (95% UI: 3.69-11.0) QALYs with standard care. Total lifetime costs were higher with tirzepatide ($131,686 [Q1-Q3: $113,005-$150,445]) vs standard care ($87,082 [Q1-Q3: $70,866-$105,268]). The resulting incremental cost-effectiveness ratio was $70,297 per QALY gained, falling below the commonly accepted United States willingness-to-pay threshold of $100,000 per QALY. Deterministic sensitivity analysis revealed that the model was most sensitive to mortality rates and health state utilities. Probabilistic sensitivity analysis found that tirzepatide was cost-effective in >50% of the simulations.

CONCLUSIONS Tirzepatide seems to be a cost-effective treatment option for patients with HFpEF and obesity in the United States. However, the modest 51% probability of cost-effectiveness underscores the need for further research and pricing adjustments to improve its economic value. (JACC Heart Fail. 2026;■:103060) © 2026 by the American College of Cardiology Foundation.

From the [a]School of Pharmacy, The University of Texas at El Paso, El Paso, Texas, USA; [b]Department of Public Health, College of Health and Human Services, North Dakota State University, Fargo, North Dakota, USA; and the [c]College of Pharmacy and Pharmaceutical Sciences, Florida A&M University, Tallahassee, Florida, USA.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

Manuscript received September 12, 2025; revised manuscript received February 12, 2026, accepted March 10, 2026.

ISSN 2213-1779/$36.00 https://doi.org/10.1016/j.jchf.2026.103060

ABBREVIATIONS AND ACRONYMS

GLP = glucagon-like peptide

HF = heart failure

HFpEF = heart failure with preserved ejection fraction

ICER = incremental cost-effectiveness ratio

KCCQ = Kansas City Cardiomyopathy Questionnaire

QALY = quality-adjusted life year

QoL = quality of life

RA = receptor agonist

SoC = standard of care

TZP = tirzepatide

wHF = worsening heart failure

Heart failure with preserved ejection fraction (HFpEF) is a growing public health concern, particularly among individuals with obesity.[1] It has been estimated that >80% of patients with HFpEF are overweight or obese.[2] Obesity contributes to the development and progression of HFpEF by promoting systemic inflammation, diastolic dysfunction, and metabolic abnormalities. In the United States, the prevalence of HFpEF is increasing in parallel with the obesity epidemic, leading to higher rates of hospitalizations, diminished quality of life (QoL), and escalating health care costs.[3] Despite the substantial clinical and economic burden, treatment options that target both HFpEF symptoms and the underlying metabolic drivers remain limited.

Tirzepatide (TZP), a novel dual agonist of glucose-dependent insulinotropic polypeptide and glucagon-like peptide (GLP)-1 receptor agonists (RAs), has shown significant benefits in weight reduction and glycemic control among patients with type 2 diabetes and obesity.[4] In the recently published SUMMIT (A Study of Tirzepatide [LY3298176] in Participants With Heart Failure With Preserved Ejection Fraction [HFpEF] and Obesity), TZP significantly reduced the risk of worsening heart failure (wHF) events and improved QoL compared with standard of care (SoC) in patients with HFpEF and obesity.[5] The SoC (ie, usual care) in the SUMMIT trial reflected background cardiovascular medications recommended by contemporary clinical guidelines, including diuretics, renin-angiotensin system and neprilysin inhibitors, and mineralocorticoid receptor antagonists.

Few prior studies have reported the cost-effectiveness of a similar class of GLP-1 RAs for the management of HF in obese patients from the U.S., Canadian, and Australian health care perspective.[6-8] However, cost-effectiveness may vary depending on drug acquisition costs and the magnitude of QoL benefits.

Although clinical evidence supports the use of TZP in this patient population, its cost and long-term economic value remain key considerations for health care decision-makers. Given the high burden of HFpEF among individuals with obesity and the potential of TZP to modify the disease trajectory, an evaluation of its cost-effectiveness is warranted. The aim of the current study was to assess the cost-effectiveness of TZP vs SoC alone in patients with HFpEF and obesity from the perspective of the U.S. health care payer, providing evidence to inform coverage and reimbursement decisions.

METHODS

POPULATION. Model inputs were derived primarily from the SUMMIT trial.[5] The SUMMIT trial is a randomized controlled trial designed to evaluate the effectiveness of TZP compared with SoC in patients with HFpEF and obesity.[5] In this trial, a total of 731 patients were randomized to treatment, with 364 patients assigned to the TZP group and 367 to the SoC group. Patients had a mean age of 65.5 ± 10.5 years and a clinical diagnosis of chronic HF, defined by NYHA functional class II to IV symptoms and a left ventricular ejection fraction of at least 50%. All participants had a body mass index ≥30 kg/m^2. Nearly one-half (46.9%) of participants had experienced a prior hospitalization or urgent care visit for wHF within the 12 months before enrollment. Overall, 80.8% of patients in the TZP group and 78.7% in the SoC group completed the trial regimen.[5]

ETHICS STATEMENT. This study used only publicly available data from published reports and did not involve human subjects' research. Therefore, Institutional Review Board approval was not required.

MODEL STRUCTURE. We developed a state-transition Markov model to evaluate the cost-effectiveness of TZP compared with SoC in adults with HFpEF and obesity. The model consisted of 3 mutually exclusive health states representing the clinical trajectory of HFpEF: stable HFpEF, wHF, and death. All patients entered the model in the stable HFpEF state. During each monthly cycle, individuals could remain in the same state, transition from stable HFpEF to wHF, or progress to death. Death was modeled as an absorbing state and stratified into cardiovascular and noncardiovascular mortality (Figure 1).

In recognition of the complexity of HFpEF, we have further developed and tested an alternative 5-health state model to compare the outcomes with our original 3-state model. The alternative 5-state Markov model was constructed based on the extent/severity of wHF and real-world management/care.[9,10] Accordingly, the 5-health state model comprised HFpEF, Outpatient care, Urgent care, Hospitalized, and Death. The 5-state model was designed to validate the findings of the original 3-state model, which is the primary modeling framework (Supplemental Figure 1).

To ensure the reliability of the model outputs, several validation steps consistent with best practices in economic modeling were performed. First, face validation was conducted through review by study team members with expertise in clinical pharmacy, epidemiology, and pharmacoeconomics. Second, internal validation was performed by evaluating whether the model produced logical results when parameter values were set to very high or very low levels (extreme-value testing). Third, cross-validation was conducted by developing an alternative 5-state model and comparing key outcomes.

To minimize bias and ensure analytic transparency, all model structures and key assumptions were prespecified before conduct of the cost-effectiveness analyses. All model inputs and assumptions followed best practices recommendations by the ISPOR-SMDM (International Society of Pharmacoeconomics and Society for Medical Decision Making) Modeling Good Research Practices Task Force.[11]

TRANSITION PROBABILITIES. Transition probabilities in the Markov model were primarily derived from treatment-specific event rates reported in the SUMMIT trial.[5] Monthly transition probabilities were calculated from the annual event rates using the standard exponential transformation formula:

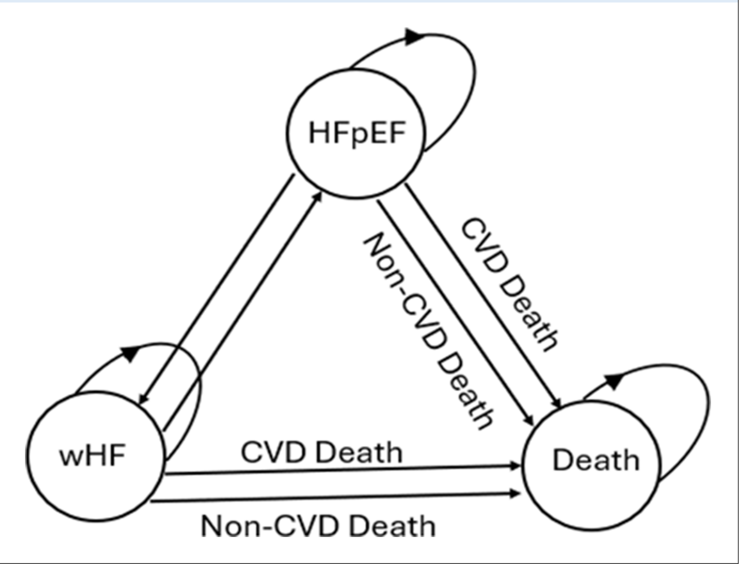
$$pmonthly = 1 - e^{-(annual\ rate/12)}$$

This method assumes a constant hazard within each monthly cycle.[12] Age-specific background mortality was incorporated using U.S. life tables to account for natural increases in mortality risk over time[13] (Table 1).

HEALTH STATE UTILITIES. Health-related QoL was assessed by using the Kansas City Cardiomyopathy Questionnaire (KCCQ) Clinical Summary Score from the SUMMIT trial. The KCCQ has been validated and is recognized as a sensitive, reproducible instrument for detecting clinical changes in HF.[14] Prior research has shown that the KCCQ is a reliable predictor of the EQ-5D-5L (EuroQol 5-Dimension, 5-Level) utility values, and mapping algorithms using the KCCQ have been widely applied in health economic evaluations.[13,15,16] A mapping technique was successfully implemented by Zheng et al[17] to estimate the cost-effectiveness of empagliflozin for the management of HFpEF.

Due to the lack of a direct measure of utility in the current cost-effectiveness model, we used a mapping algorithm by applying an ordinary least squares regression method to convert KCCQ to EQ-5D-5L values. The mapping algorithm/function was developed from previously published studies.[18,19]

FIGURE 1 Schematic of the Markov Model for HFpEF and Obesity



This figure illustrates the Markov model structure used to simulate disease progression. Arrows indicate possible transitions between health states in each cycle of the model. CVD = cardiovascular disease; HFpEF = heart failure with preserved ejection fraction; wHF = worsening heart failure.

Different states of QoL were assigned for patients with HFpEF and wHF health states.[20] The QoL for these states was sourced from the TEN-HMS (Trans-European Network Home-Care Management System) trial.[21] The monthly utilities values of each health state are presented in Table 2.

COST INPUTS AND DISCOUNTS. The model incorporated direct medical costs from a U.S. health care payer perspective, including drug acquisition, routine medical management, and treatment of adverse events. All costs were adjusted to 2025 U.S. dollars using the medical care component of the Consumer Price Index.[22] Both costs and health outcomes were discounted at an annual rate of 3%. A 20-year lifetime horizon was used to capture the long-term costs and outcomes of treatment.

TABLE 1 Monthly Transition Probabilities With 95% Range

Transition	Tirzepatide	SoC	Trial Name
HFpEF → wHF	0.0038 (0.0029-0.0048)	0.0068 (0.0051-0.0085)	SUMMIT
HFpEF → death	0.0024 (0.0018-0.0030)	0.0018 (0.0014-0.0023)	SUMMIT
Remain in HFpEF state	0.9938 (0.7453-1.0000)	0.9914 (0.7435-1.0000)	SUMMIT
wHF → death	0.00069 (0.0005-0.0009)	0.00023 (0.0002-0.0003)	SUMMIT
Remain in wHF state	0.99571 (0.7468-1.0000)	0.9991 (0.7471-1.0000)	SUMMIT
wHF → HFpEF	0.0036 (0.0027-0.0045)	0.0037 (0.0028-0.0046)	SUMMIT

HFpEF = heart failure with preserved ejection fraction; SoC = standard of care; SUMMIT = A Study of Tirzepatide (LY3298176) in Participants With Heart Failure With Preserved Ejection Fraction (HFpEF) and Obesity; wHF = worsening heart failure.

TABLE 2 Monthly Health State Utility Values in Patients With HFpEF and Obesity

	Monthly Utility	SoC	First Author
HFpEF	0.0625 (0.0467-0.0783)	0.0550 (0.0408-0.0692)	Packer et al[5]
wHF	0.0525 (0.0433-0.0717)	0.0450 (0.0367-0.0617)	Johansson et al Di Tanna et al[20] de Almeida et al[21]

Johansson I, Joseph P, Balasubramanian K, et al. Health-related quality of life and mortality in heart failure: the global congestive heart failure study of 23 000 patients from 40 countries. *Circulation*. 2021;143(22):2129-2142.

Abbreviations as in Table 1.

A half-cycle correction factor was applied to both costs and effectiveness.

The monthly cost of TZP was modeled based on the recommended dose escalation schedule used in the SUMMIT trial.[23] Total TZP costs resulted in an unadjusted annual cost of $10,338 per patient. Acquisition costs for TZP were further adjusted to account for a 37% rebate, consistent with previously reported estimates for U.S. payers. Specifically, Tang et al[24] applied a 37% rebate in their long-term cost-effectiveness analysis of TZP, and historical data indicate that GLP-1 RA rebates accounted for approximately 37% of total payments from a payer perspective.[25] This rebate resulted in annual drug cost of $6,522.94. Further adjustment accounted for a treatment discontinuation rate, resulting in a weighted annual cost of $5,258.64, assuming that patients who discontinue treatment no longer incur drug costs.

To capture the impact of long-term weight outcomes, weight-regain/change assumptions were incorporated into the model by extrapolating from semaglutide evidence, as TZP-specific post-withdrawal/discontinuation data are not yet available. In the STEP 1 (Research Study Investigating How Well Semaglutide Works in People Suffering From Overweight or Obesity) extension study, semaglutide users regained approximately 60% to 70% of previously lost weight at 1 year.[26] These regain rates were applied to the 52-week percent weight changes observed in TZP and SoC in the SUMMIT trial.[5] To estimate downstream economic implications of this weight change, we used published evidence from Thorpe and Joski[27] indicating that a 5% weight change/loss yields $670 in annual health care savings; these values were incorporated as negative cost terms in TZP health states.

Costs associated with wHF were stratified according to care setting. For patients requiring intravenous diuretic therapy in an urgent care setting, costs were derived from the treatment algorithm outlined by Buckley et al,[28] in which patients receive a 3-hour infusion of intravenous furosemide. For those requiring outpatient intensification with oral diuretics, costs were estimated based on a maximum oral furosemide dose of 600 mg/day. The model also incorporated common TZP-associated adverse events and key comorbidities that influence HFpEF outcomes. Adverse events included diarrhea, constipation, vomiting/nausea, dyspepsia, loss of appetite, dizziness/hypotension, and abdominal pain. Comorbid conditions (ie, diabetes mellitus) were included as events with associated costs when appropriate.[29] A full description of costs and their components is presented in **Table 3**, with additional details on costs and other model inputs provided in the Supplemental Methods (Supplemental Table 1).

OUTCOMES AND ANALYSES. The primary outcome was the incremental cost-effectiveness ratio (ICER), expressed as cost per quality-adjusted life years (QALYs) gained. Secondary outcomes included total costs and total QALYs. One-way sensitivity analyses were performed on key model parameters to identify drivers of model uncertainty. In the absence of empirical measures of uncertainty (eg, SEs or CIs), sensitivity analyses were conducted by varying values by ±25% from their base-case values.[30,31]

Probabilistic sensitivity analysis using Monte Carlo simulation (10,000 iterations) was conducted to assess the likelihood that TZP is cost-effective at a willingness-to-pay (WTP) threshold of $100,000 per QALY, consistent with the U.S. gross domestic product per capita.[24,32,33] Additional scenario analyses were conducted focusing on mortality assumptions. In scenario 1, we assumed no mortality effect/benefit of TZP. In scenario 2, we applied the mortality HR reported in the SUMMIT trial to adjust the TZP death probabilities.

The analysis was conducted by using a cohort state-transition model parameterized with aggregate trial-level inputs; outcomes are reported on a per-patient basis on a 20-year time horizon and do not reflect analyses of individual patient-level data. The study findings are reported in accordance with the Consolidated Health Economic Evaluation Reporting Standards Statement.[34] The analysis was implemented by using R version 4.5.0 and constructed with the heemod package.[35,36]

RESULTS

BASE-CASE ANALYSIS. In the base-case analysis, treatment with TZP was associated with increased QALYs compared with SoC alone. Patients receiving TZP gained an average of 7.27 (95% uncertainty

TABLE 3 Monthly Cost Estimates for the Management of Heart Failure

	Costs, USD	First Author or Title
Tirzepatide (drug acquisition)	438.53 (328.90-548.17)	Lilly[23]
Heart failure hospitalization	818.18 (613.64-1,022.73)	Urbich et al
Diabetes treatment	1,157.38 (868.03-1,446.73)	Parker et al[29]
Urgent intravenous diuretic use	60.00 (45.71-75.71)	Buckley et al,[28] Bensimhon et al
Oral diuretic intensification	13.8 (10.4-17.2)	Wu et al, McDonagh et al
Regular dose/use of diuretics	0.95 (0.72-1.20)	Wu et al, McDonagh et al
Renin-angiotensin system inhibitors	2.60 (1.95-3.25)	Lisinopril prices and coupons
Beta-blockers	2.99 (2.24-3.73)	Metoprolol prices, coupons, and savings tips
Mineralocorticoid receptor antagonists	3.00 (2.25-3.75)	Spironolactone prices and coupons
Sodium-glucose cotransporter 2 inhibitors	353.1 (265.9-441.4)	Spironolactone prices and coupons
Adverse drug reaction costs		
Diarrhea	258.3 (193.7-322.8)	Buja et al, Price for loperamide
Constipation	107.3 (80.4-134.0)	Pekmezaris et al
Vomiting/nausea	185.4 (139.0-231.7)	Gress et al
Dyspepsia	300.0 (225.0-375.0)	Lacy et al
Loss of appetite	3,000.0 (2,250.0-3,750.0)	Streatfeild et al
Urinary tract infection	9,800.0 (7,350.0-12,250.0)	Simmering et al
Atrial fibrillation (hospital)	14,000.0 (10,500.0-17,500.0)	Buja et al
Cardiac failure (hospital)	13,600.0 (10,200.0-17,000.0)	Osenenko et al
Anemia (hospital)	31,250.0 (23,440.0-39,060.0)	Nissenson et al
Dizziness/hypotension	739.5 (555.2-925.0)	Kovacs et al
Abdominal pain	1,886.2 (1,414.7-2,357.8)	Li et al

Values are median (Q1-Q3). Bensimhon D, Weintraub WS, Peacock WF, et al. Reduced heart failure-related healthcare costs with Furoscix versus in-hospital intravenous diuresis in heart failure patients: the FREEDOM-HF study. *Future Cardiol*. 2023;19(8):385-396. Buja A, Rebba V, Montecchio L, et al. The cost of atrial fibrillation: a systematic review. *Value Health*. 2024;27(4):527-541. Gress K, Urits I, Viswanath O, Urman RD. Clinical and economic burden of postoperative nausea and vomiting: analysis of existing cost data. *Best Pract Res Clin Anaesthesiol*. 2020;34(4):681-686. Kovacs E, Wang X, Grill E. Economic burden of vertigo: a systematic review. *Health Econ Rev*. 2019;9(1):37. Lacy B, Weiser K, Kennedy A, Crowell M, Talley N. Functional dyspepsia: the economic impact to patients. *Aliment Pharmacol Ther*. 2013;38(2):170-177. Li R, Tham SW, Basu A, Palermo TM, Dang TT, Groenewald CB. Medical expenditures associated with abdominal and pelvic pain in the United States, 2017-2021. *J Pain*. 2025;28:104793. Lisinopril prices and coupons. GoodRx. Accessed August 4, 2025. https://www.goodrx.com/lisinopril. Metra M, Adamo M, et al. 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2021;42(36):3599-3726. Metoprolol prices, coupons, and savings tips. GoodRx. Accessed August 4, 2025. https://www.goodrx.com/metoprolol. Nissenson AR, Wade S, Goodnough T, Knight K, Dubois RW. Economic burden of anemia in an insured population. *J Manag Care Pharm*. 2005;11(7):565-574. Osenenko KM, Kuti E, Deighton AM, Pimple P, Szabo SM. Burden of hospitalization for heart failure in the United States: a systematic literature review. *J Manag Care Specialty Pharm*. 2022;28(2):157-167. Pekmezaris R, Aversa L, Wolf-Klein G, Cedarbaum J, Reid-Durant M. The cost of chronic constipation. *J Am Med Dir Assoc*. 2002;3(4):224-228. Price for loperamide. GoodRx. Accessed July 17, 2025. https://www.goodrx.com/loperamide. Simmering JE, Tang F, Cavanaugh JE, Polgreen LA, Polgreen PM. The increase in hospitalizations for urinary tract infections and the associated costs in the United States, 1998-2011. *Open Forum Infect Dis*. 2017;4(1):ofw281. Spironolactone prices and coupons. GoodRx. Accessed August 4, 2025. https://www.goodrx.com/spironolactone. Streatfeild J, Hickson J, Austin SB, et al. Social and economic cost of eating disorders in the United States: evidence to inform policy action. *Int J Eating Dis*. 2021;54(5):851-868. Urbich M, Globe G, Pantiri K, et al. A systematic review of medical costs associated with heart failure in the USA (2014-2020). *Pharmacoeconomics*. 2020;38(11):1219-1236. Wu L, Rodriguez M, El Hachem K, Krittanawong C. Diuretic treatment in heart failure: a practical guide for clinicians. *J Clin Med*. 2024;13(15):4470.

USD = United States dollars.

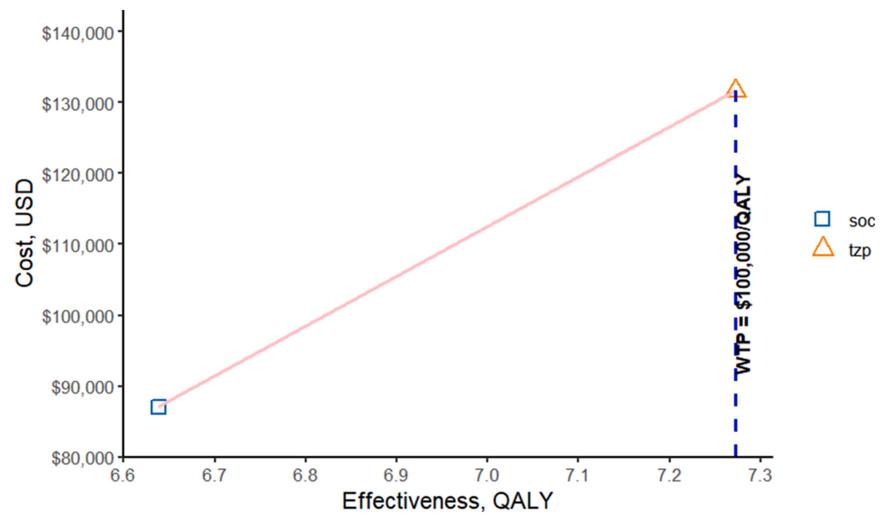
interval [UI]: 3.20-13.7) QALYs vs 6.63 (95% UI: 3.69-11.0) QALYs in the SoC group. Total lifetime costs were higher with TZP ($131,686 [Q1-Q3: $113,005-$150,445]) vs SoC ($87,082 [Q1-Q3: $70,866-$105,268]). The resulting ICER was $70,297 per QALY gained, which is below the commonly accepted WTP threshold in the United States ($100,000 per QALY) (Figure 2, Table 4).

In addition, the alternative 5-state model produced results that were consistent with the findings of our original model. Specifically, TZP generated a total lifetime cost of $117,175.45 and 6.58 QALYs, compared with $72,531.22 and 6.05 QALYs for the SoC. These values correspond to an ICER of $84,781 per QALY gained, which remains below the commonly accepted U.S. WTP threshold of $100,000 per QALY. These results show that even after incorporating additional clinical complexity into the disease pathway, TZP remains cost-effective.

The mortality-based scenario analyses reported that, in the no-mortality-effect scenario, TZP had a lifetime cost of $138,339 compared with $87,082 for SoC, producing 7.649 QALYs vs 6.638 QALYs for SoC. This resulted in an incremental cost of $51,258, 1.011 incremental QALYs, and an ICER of $50,701 per QALY gained. These analyses indicate that the costs and QALY gains of TZP were sensitive to mortality assumptions but that the ICER remained below commonly cited U.S. WTP thresholds.

DETERMINISTIC SENSITIVITY ANALYSIS. The one-way deterministic sensitivity analysis examined key

FIGURE 2 Cost-Effectiveness of TZP vs SoC for the Management of HFpEF and Obesity



This figure presents a cost-effectiveness plane comparing tirzepatide (TZP) with standard of care (SoC) over a 20-year horizon. Costs are shown on the y-axis, and effectiveness in quality-adjusted life year (QALY) is shown on the x-axis. A diagonal line connects the SoC point to the TZP point, illustrating the incremental cost and effectiveness of TZP relative to SoC. WTP = willingness-to-pay; other abbreviation as in Figure 1.

parameters to assess the robustness of the base-case cost-effectiveness results. The model was most sensitive to assumptions about health utility values and mortality risks in both treatment arms, which produced the widest range of ICERs. Increasing the TZP utility value by 25% decreased the ICER below the WTP threshold and made the TZP strategy cost-effective, whereas increasing the SoC utility by 25% produced ICER ranges beyond the WTP threshold of $100,000 per QALY gain. Changes in mortality risk generated similarly large movements. When the mortality probability for the TZP arm increased by 25%, the ICER rose to $117,535 per QALY. Conversely, increasing the mortality in the SoC arm lowered the ICER to $57,204 per QALY. Changes in the cost parameters, discount rate, and transition probabilities had relatively moderate to small impacts on the ICERs tested values (Figure 3).

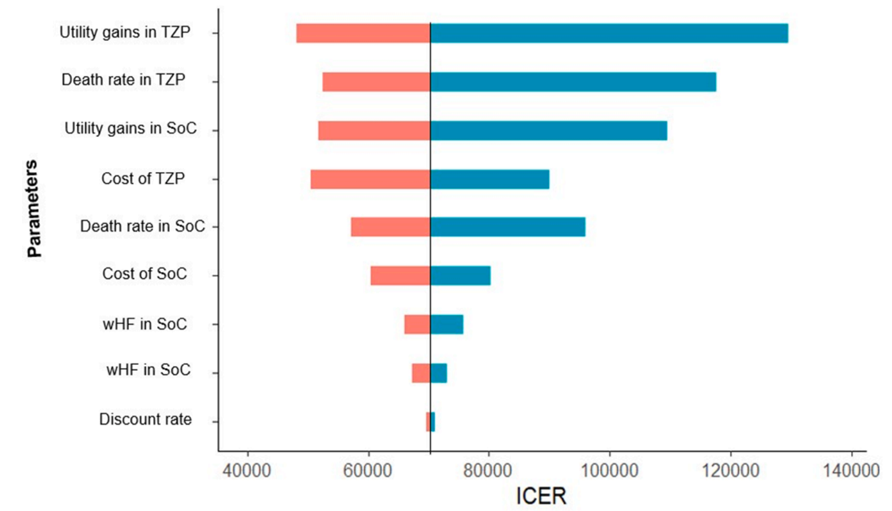
TABLE 4 Base-Case Cost-Effectiveness Results For TZP vs SoC

	Base-Case Value
Total cost TZP	$131,686 (Q1-Q3: $113,005-$150,445)
Total QALY TZP	7.27 (95% UI: 3.20-13.7)
Incremental cost TZP vs SoC	44,604
Incremental QALY TZP vs SoC	0.64
Total cost SoC	$87,082 (Q1-Q3: $70,866-$105,268)
Total QALY SoC	6.63 (95% UI: 3.69-11.0)
ICER	$70,297 per QALY

ICER = incremental cost-effectiveness ratio; QALY = quality-adjusted life year; TZP = tirzepatide; UI = uncertainty interval; other abbreviation as in Table 1.

PROBABILISTIC SENSITIVITY ANALYSIS. A probabilistic sensitivity analysis was conducted by using 10,000 Monte Carlo simulations to characterize the uncertainty surrounding model parameters. Across all iterations, TZP was associated with higher mean total costs and QALYs compared with SoC. The mean cost per patient was $87,377.92 for SoC and $132,060.41 for TZP; the mean QALYs gained were 6.64 and 7.37, respectively. The cost-effectiveness scatterplot illustrates the distribution of ICERs in relation to a WTP threshold of $100,000 per QALY (Figure 4). The cost-effectiveness acceptability curve shows the probability of cost-effectiveness across a range of WTP thresholds. The SoC strategy had a high probability of cost-effectiveness at lower WTP values, which declined gradually and dropped below 50% at approximately $100,000 per QALY. In contrast, the probability that TZP is cost-effective increased with rising WTP thresholds, surpassing SoC at the $100,000 per QALY mark. The probability that TZP was cost-effective at WTP thresholds of $50,000, $100,000, $200,000, and $300,000 per QALY was 45.5%, 51.4%, 53.3%, and 54.5%, respectively (Figure 5).

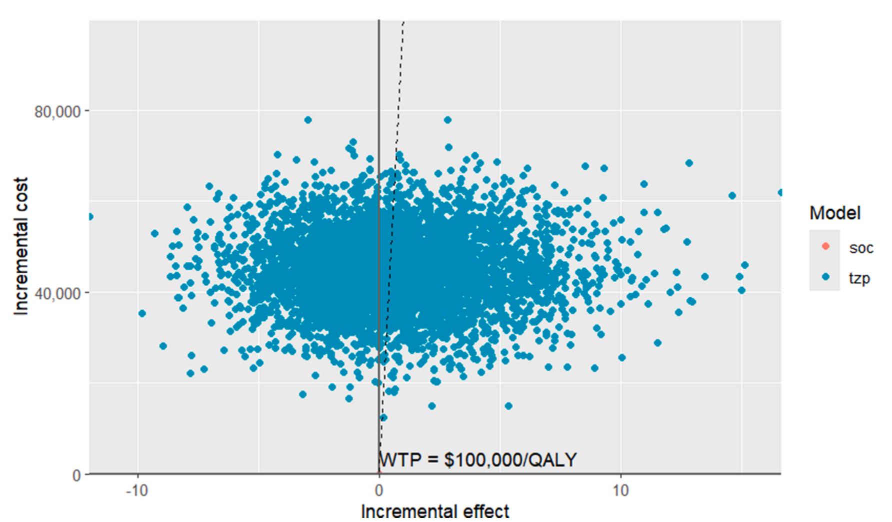
FIGURE 3 **Key Drivers of Cost-Effectiveness From Deterministic Sensitivity Analysis**



This figure presents a tornado diagram illustrating the key parameters influencing the cost-effectiveness of TZP vs SoC in heart failure management. Parameters with longer bars have a greater effect on the model's results, highlighting the main drivers of uncertainty. ICER = incremental cost-effectiveness ratio; other abbreviations as in Figures 1 and 2.

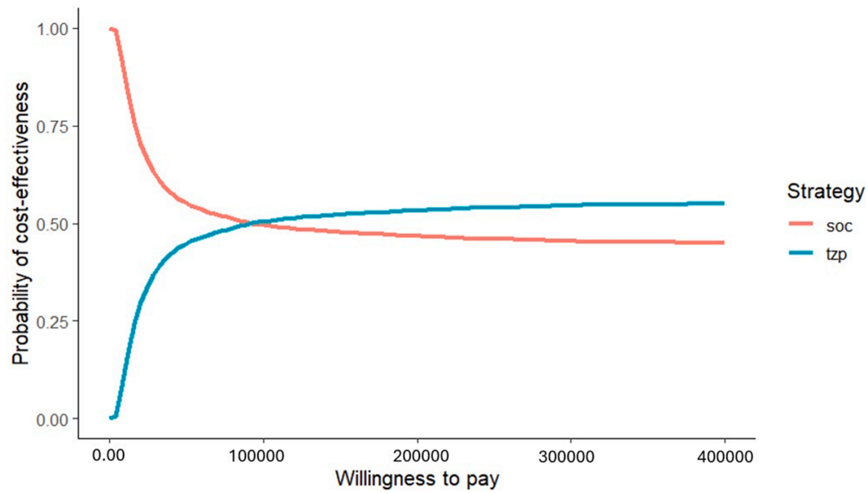
LONG-TERM CLINICAL BENEFIT OF TZP VS SoC FOR PATIENTS WITH HFpEF AND OBESITY. The projected probability of patients occupying each health state over a 20-year horizon was analyzed. Figure 6, top, shows that cumulative mortality rises steadily in both treatment arms, with TZP slightly higher than SoC, suggesting limited survival advantage. Figure 6, middle, indicates that patients receiving TZP have a higher probability of remaining in a stable HFpEF state throughout the model horizon, reflecting slower disease progression and improved clinical stability. Figure 6, bottom, shows that the probability

FIGURE 4 **Cost-Effectiveness Plane Scatter Plot of TZP for Heart Failure Management**



This figure shows the cost-effectiveness plane comparing TZP with SoC over a 20-year horizon. Each point represents a simulation from the probabilistic sensitivity analysis, plotting incremental costs against QALYs. Abbreviations as in Figure 2.

FIGURE 5 **Cost-Effectiveness Acceptability Curve for TZP vs SoC**



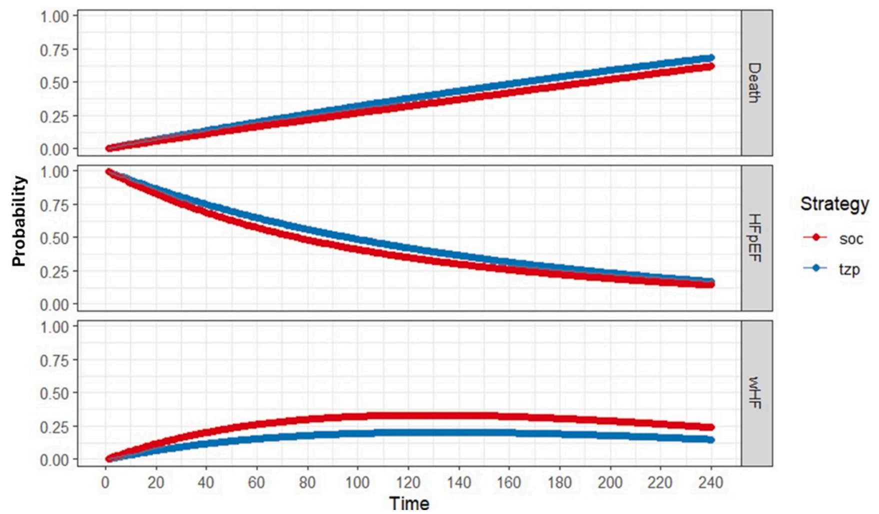
This figure illustrates the cost-effectiveness acceptability curve comparing TZP and SoC for heart failure management. The curve shows the probability that each treatment strategy is cost-effective across a range of WTP thresholds. Abbreviations as in Figure 2.

of entering the wHF state increases more slowly in the TZP arm than in SoC, suggesting that TZP may reduce or delay transitions to more severe HF stages, potentially leading to better QoL over time. The overall model framework and key cost-effectiveness outcomes are illustrated in the Central Illustration.

DISCUSSION

This study evaluated the long-term cost-effectiveness of TZP for the management of HFpEF in patients with obesity from the U.S. health care payer perspective. TZP was associated with improved

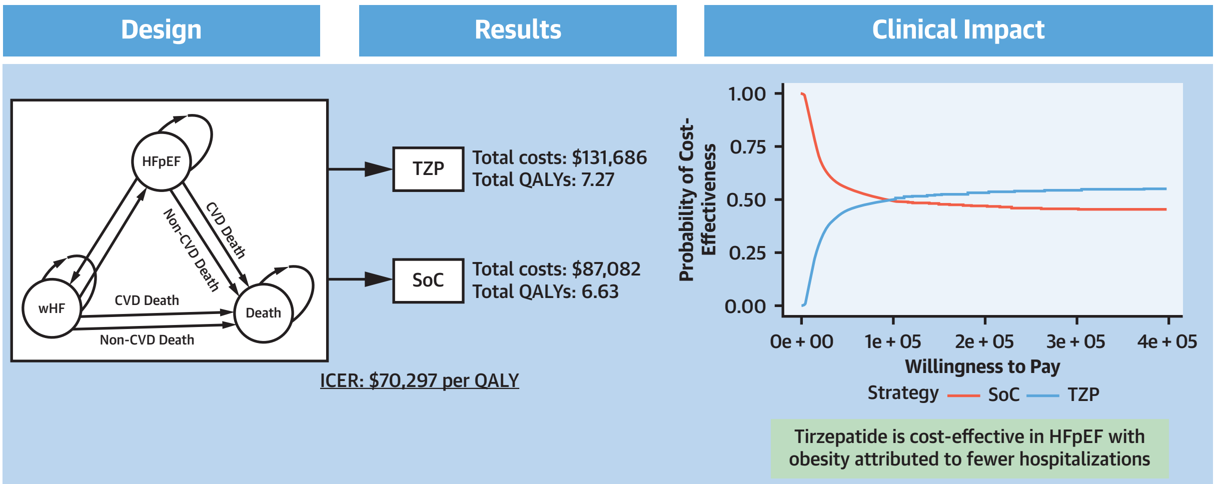
FIGURE 6 **Long-Term Clinical Benefit of TZP For Patients With HFpEF and Obesity**



This figure illustrates the projected long-term clinical outcomes for patients with HFpEF and obesity treated with TZP vs SoC over the 20-year model horizon, including cumulative mortality, probability of remaining in stable HFpEF, and probability of worsening heart failure. Abbreviations as in Figures 1 and 2.

ARTICLE IN PRESS

CENTRAL ILLUSTRATION **Overview of Cost-Effectiveness of Tirzepatide in Patients With HFpEF and Obesity**



Abegaz TM, et al. JACC Heart Fail. 2026;■(■):103060.

This illustration summarizes the overall economic benefits of tirzepatide (TZP) for patients with heart failure with preserved ejection fraction (HFpEF) and obesity. CVD = cardiovascular disease; HFpEF = heart failure with preserved ejection fraction; ICER = incremental cost-effectiveness ratio; QALY = quality-adjusted life year; SoC = standard of care; wHF = worsening heart failure.

clinical outcomes, yielding a gain of 0.64 QALY compared with SoC, albeit at a higher cost. The resulting ICER of $70,297 per QALY falls below the commonly accepted U.S. WTP threshold of $100,000 per QALY, suggesting that TZP may be a cost-effective intervention in this population.

Our findings are consistent with prior economic evaluations of GLP-1 RAs, which have shown favorable cost-effectiveness profiles in populations with obesity and cardiovascular disease. For instance, the GLP-1 RA semaglutide was found to be cost-effective in a Canadian health care setting, with an ICER of $72,962 per QALY gained; however, the probability of cost-effectiveness was only 14%, largely due to the use of a lower WTP threshold of $50,000 per QALY.[7] Another cost-effectiveness analysis of semaglutide in individuals with obesity and established cardiovascular disease without diabetes reported a higher ICER of $136,271 per QALY at the list price.[6] The elevated ICER in that study may be attributed to the exclusion of a drug rebate, whereas the current analysis applied a 37% rebate. Furthermore, a separate evaluation of semaglutide for secondary prevention of cardiovascular disease in the United States reported an ICER of $443,000 per QALY gained and recommended a 79% price reduction to meet the $100,000 per QALY cost-effectiveness threshold.[37] Collectively, these findings underscore the importance of drug pricing in determining the cost-effectiveness of GLP-1 RAs and highlight the need for pricing strategies that align with value-based care principles.

The current deterministic sensitivity analysis showed that health-state utility values and mortality probabilities were the principal drivers of uncertainty in the model. This is likely because patients spend many years in the stable HFpEF state, meaning that even small changes in the utility assigned to this state resulted in meaningful differences in lifetime QALYs. Consequently, these variations produced wide swings in ICER estimates across the tested input ranges.[38] Mortality assumptions similarly exerted substantial influence. Variations in death rates in both the TZP and SoC arms generated the largest shifts in the ICER, reflecting the direct relationship between survival duration, accumulated QALYs, and total costs. Because survival duration directly determines the amount of time individuals accumulate QALYs and incur health care costs, variations in death rates altered both the total benefits and total expenditures.[39]

TZP has been reported to improve QoL and reduce progression to worse health states. The SUMMIT trial showed that patients who received TZP experienced improvements in QoL, reflecting the utility benefit of TZP over SoC.[5] In addition, the SURPASS clinical trial program, which assessed the efficacy and safety of

TZP for type 2 diabetes, reported significant improvements in health- and weight-related QoL comparisons.[40] Furthermore, the SURMOUNT-1 (A Study of Tirzepatide [LY3298176] in Participants With Obesity or Overweight) trial, which investigated the effects of TZP on body weight in individuals with obesity or overweight, found that TZP treatment was associated with improved QoL compared with placebo.[41] The improved QoL in the TZP group may be partially attributed to greater weight reduction, making TZP the most pragmatic therapy for patients with HFpEF comorbid with overweight/obesity. Nonetheless, the long-term QoL benefit may diminish due to the accumulation of deaths over time, as TZP has not shown superiority in reducing mortality compared with SoC.[5]

Probabilistic sensitivity analysis revealed that TZP was cost-effective in >50% of simulations at a WTP threshold of $100,000 per QALY, reflecting moderate uncertainty. The cost-effectiveness acceptability curve showed that TZP becomes increasingly likely to be cost-effective at higher WTP thresholds, with probabilities estimated to be 53.3% at thresholds above $200,000 per QALY. These results reflect a value proposition that may appeal to payers and decision-makers, particularly in settings in which higher thresholds are acceptable due to the severity of illness and lack of alternative treatments.[42] In interpreting the probabilistic sensitivity analysis results, it is important to note that all structural assumptions and base-case parameter inputs were prespecified before analysis. This prespecification mitigates the risk of selective modeling choices influencing the results.

The current analysis showed that the inclusion of long-term weight regain and obesity-related comorbidity progression significantly increases lifetime medical costs, which in turn, influences the resulting ICER.[43] Therefore, accurately accounting for long-term weight change and obesity-related conditions is essential when developing economic evaluation models of antiobesity therapies.

STUDY STRENGTHS AND LIMITATIONS. First, this study was one of the first cost-effectiveness evaluations focused specifically on TZP for HFpEF in patients with obesity. Second, the model structure was informed by rigorous clinical trial data, which increases its credibility and relevance. Third, both deterministic and probabilistic sensitivity analyses were conducted comprehensively, enhancing the robustness of the results. Lastly, the study followed best practices modeling guidelines, including adherence to the ISPOR-SMDM Modeling Good Research Practices framework, ensuring transparency and reproducibility of the findings.

However, the following limitations should be acknowledged. First, it is widely acknowledged that projecting lifetime costs and QALYs beyond the trial horizon introduces inherent uncertainty and the potential for bias. Second, the SUMMIT trial did not collect preference-based QoL measures, requiring us to estimate utilities by mapping KCCQ scores to EQ-5D-5L values. Although this approach is consistent with prior economic evaluations, mapping introduces uncertainty because the prediction algorithms are developed from external cohorts and may not fully capture treatment-related QoL improvements. Consequently, the magnitude of QALY gains may be either overestimated or underestimated, and the ICER estimates may be sensitive to potential bias and uncertainty associated with the mapping process.[17] Future studies incorporating directly elicited utilities, such as EQ-5D-5L or the SF-6D (Short-Form 6-Dimension), would substantially strengthen the evidence base and reduce uncertainty surrounding health state valuations. In addition, assuming constant transition probabilities may oversimplify the dynamic progression of HFpEF and the time-varying effects of treatment; however, this approach is consistent with prior cost-effectiveness analyses and reflects common practice when long-term, time-varying transition data are unavailable. The findings reflect average effects over the analytic horizon rather than time-specific risk estimates. Moreover, a broader societal perspective, including indirect costs such as productivity loss and caregiver burden, was not included due to limited data. As a result, the full economic impact of the interventions may be underestimated. Future evaluations should consider wider perspectives to capture these indirect costs. Finally, the use of individual-level data would have allowed for more detailed assessment of heterogeneity in costs and outcomes compared with aggregate-level inputs; such data, however, were not feasible to obtain.

FUTURE DIRECTIONS. Future research should aim to validate and refine the cost-effectiveness of TZP using long-term real-world evidence and observational data beyond the clinical trial setting. Although the current study relied on extrapolated projections from the SUMMIT trial, extended follow-up data surveillance could offer more accurate estimates of long-term clinical outcomes, treatment adherence, and real-world health care resource utilization. In addition, subpopulation analyses from patient-level data based on demographic or clinical characteristics may help identify patient groups most likely to benefit from TZP.

CONCLUSIONS

This cost-effectiveness analysis suggests that TZP is a potentially cost-effective treatment option for patients with HFpEF and obesity in the United States when evaluated using a 3-state Markov model that reflects the primary clinical outcomes of interest. Although associated with higher upfront costs, TZP provided meaningful gains in QALYs and reduced wHF events, resulting in an ICER of $70,297 per QALY, below the commonly accepted U.S. WTP of $100,000 per QALY. The use of the expanded 5-state model design to validate the robustness of the primary modeling framework better reflected the clinical pathways and yielded consistent findings. Payers and health care decision-makers may consider the inclusion of TZP as a cost-effective treatment option for eligible patients with HFpEF and obesity.

DATA AVAILABILITY The authors confirm that the data supporting the findings of this study are available within the paper and the Supplemental Appendix.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

The authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Tadesse M. Abegaz, School of Pharmacy, The University of Texas at El Paso, 1101 North Campbell Street, El Paso, Texas 79902, USA. E-mail: tabegaz@utep.edu.

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: TZP improves QoL and reduces progression to wHF in patients with HFpEF and obesity, with an ICER of $70,297 per QALY. Cost-effectiveness is most sensitive to mortality rates and utility assumptions.

TRANSLATIONAL OUTLOOK: Real-world studies are needed to validate the long-term cost-effectiveness of TZP. Economic evaluations incorporating societal costs may better capture the value of TZP. Subgroup analyses could help identify patient groups most likely to benefit.

REFERENCES

1. Borlaug BA, Jensen MD, Kitzman DW, Lam CS, Obokata M, Rider OJ. Obesity and heart failure with preserved ejection fraction: new insights and pathophysiological targets. *Cardiovasc Res*. 2022;118(18):3434-3450.

2. Lee DY. Obesity and heart failure with preserved ejection fraction: pathophysiology and clinical significance. *Cardiovasc Prevent Pharmacother*. 2022;4(2):70-74.

3. Golla MSG, Shams P. *Heart Failure With Preserved Ejection Fraction (HFpEF)*. StatPearls Publishing; 2024.

4. Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med*. 2022;387(3):205-216.

5. Packer M, Zile MR, Kramer CM, et al. Tirzepatide for heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2025;392(5):427-437.

6. McEwan P, Bøg M, Faurby M, et al. Cost-effectiveness of semaglutide in people with obesity and cardiovascular disease without diabetes. *J Med Econ*. 2025;28(1):268-278.

7. Rennert-May E, Manns B, Clement F, et al. Cost-effectiveness of semaglutide in patients with obesity and cardiovascular disease. *Can J Cardiol*. 2025;41(1):128-136.

8. Zomer E, Zhou J, Nelson AJ, et al. The cost-effectiveness of semaglutide in reducing cardiovascular risk among people with overweight and obesity and existing cardiovascular disease, but without diabetes. *Eur Heart J Qual Care Clin Outcomes*. 2025;11(6):857-867.

9. Khan MS, Butler J, Greene SJ. Recognizing the significance of outpatient worsening heart failure. *J Am Heart Assoc*. 2020;9(14):e017485.

10. Madelaire C, Gustafsson F, Stevenson LW, et al. One-year mortality after intensification of outpatient diuretic therapy. *J Am Heart Assoc*. 2020;9(14):e016010.

11. Caro JJ, Briggs AH, Siebert U, Kuntz KM. Modeling good research practices—overview: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force-1. *Medical Decision Making*. 2012;32(5):667-677.

12. Spruance SL, Reid JE, Grace M, Samore M. Hazard ratio in clinical trials. *Antimicrob Agents Chemother*. 2004;48(8):2787-2792.

13. Spertus J, Peterson E, Conard MW, et al. Monitoring clinical changes in patients with heart failure: a comparison of methods. *Am Heart J*. 2005;150(4):707-715.

14. Pokharel Y, Khariton Y, Tang Y, et al. Association of serial Kansas City Cardiomyopathy Questionnaire assessments with death and hospitalization in patients with heart failure with preserved and reduced ejection fraction: a secondary analysis of 2 randomized clinical trials. *JAMA Cardiol*. 2017;2(12):1315-1321.

15. Thomas M, Jones PG, Cohen DJ, et al. Predicting the EQ-5D utilities from the Kansas City Cardiomyopathy Questionnaire in patients with heart failure. *Eur Heart J Qual Care Clin Outcomes*. 2021;7(4):388-396.

16. Kaufman BG, Granger BB, Sun J-L, et al. The cost-effectiveness of palliative care: insights from the PAL-HF trial. *J Cardiac Fail*. 2021;27(6):662-669.

17. Zheng J, Parizo JT, Spertus JA, Heidenreich PA, Sandhu AT. Cost-effectiveness of empagliflozin in patients with heart failure with preserved ejection fraction. *JAMA Intern Med*. 2022;182(12):1278-1288.

18. Hunger M, Eriksson J, Regnier SA, Mori K, Spertus JA, Cristino J. Mapping the Kansas City Cardiomyopathy Questionnaire (KCCQ) onto EQ-5D-3L in heart failure patients: results for the Japanese and UK value sets. *MDM Policy Pract*. 2020;5(2):2381468320971606.

19. Ravera A, Santema BT, Sama IE, et al. Quality of life in men and women with heart failure: association with outcome, and comparison between the Kansas City Cardiomyopathy Questionnaire and the EuroQol 5 dimensions questionnaire. *Eur J Heart Fail*. 2021;23(4):567-577.

20. Di Tanna GL, Urbich M, Wirtz HS, et al. Health state utilities of patients with heart failure: a systematic literature review. *Pharmacoeconomics*. 2021;39(2):211-229.

21. de Almeida FA, Maiwenn J, Al, Koymans R, Riistama J, Pauws S, Severens JL. Impact of hospitalisation on health-related quality of life in patients with chronic heart failure. *Health Qual Life Outcomes*. 2020;18(1):262.

22. Consumer Price Index. U.S. Bureau of Labor Statistics. Accessed July 21, 2025. https://www.bls.gov/cpi/

23. How much should I expect to pay for Mounjaro (tirzepatide). Lilly. Accessed July 11, 2025. https://pricinginfo.lilly.com/mounjaro

24. Tang M, Shi L, Guan D, et al. Long-term cost-effectiveness of tirzepatide for individuals with type 2 diabetes and comorbid obesity. *Diabetes Obes Metab*. 2025;27(8):4288–4299.

25. Ding Y, Shi L, Lin Y, Shao H. 132-OR: trends in total payments, net payments, and rebates for a 30-day supply of glucose-lowering drugs (GLDs) from 2010–2017. *Diabetes*. 2022;71(suppl 1): 132-OR.

26. Wilding JP, Batterham RL, Davies M, et al. Weight regain and cardiometabolic effects after withdrawal of semaglutide: the STEP 1 trial extension. *Diabetes Obes Metab*. 2022;24(8): 1553–1564.

27. Thorpe KE, Joski PJ. Estimated reduction in health care spending associated with weight loss in adults. *JAMA Network Open*. 2024;7(12): e2449200.

28. Buckley LF, Carter DM, Matta L, et al. Intravenous diuretic therapy for the management of heart failure and volume overload in a multidisciplinary outpatient unit. *JACC Heart Fail*. 2016;4(1):1–8.

29. Parker ED, Lin J, Mahoney T, et al. Economic costs of diabetes in the U.S. in 2022. *Diabetes Care*. 2024;47(1):26–43.

30. Briggs AH, Weinstein MC, Fenwick EA, et al. Model parameter estimation and uncertainty: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force-6. *Value Health*. 2012;15(6):835–842.

31. Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. *Methods for the Economic Evaluation of Health Care Programmes*. 4th ed. Oxford, United Kingdom: Oxford University Press; 2015.

32. Anderson JL, Heidenreich PA, Barnett PG, et al. ACC/AHA statement on cost/value methodology in clinical practice guidelines and performance measures: a report of the American College of Cardiology/American Heart Association Task Force on Performance Measures and Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2014;63(21):2304–2322.

33. Robinson LA, Hammitt JK, Chang AY, Resch S. Understanding and improving the one and three times GDP per capita cost-effectiveness thresholds. *Health Policy Planning*. 2017;32(1):141–145.

34. Husereau D, Drummond M, Augustovski F, et al. Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) statement: updated reporting guidance for health economic evaluations. *MDM Policy Practice*. 2022;7(1):23814683211061097.

35. R Core. Team. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Accessed April 12, 2026. https://www.R-project.org/

36. Filipović-Pierucci A, Zarca K, Durand-Zaleski I. Markov models for health economic evaluation: the R package heemod. *arXiv*. Published online April 25, 2017. https://doi.org/10.48550/arXiv.1702.03252

37. Hennessy S, Penko J, Bellows B, et al. Cost-effectiveness of semaglutide for secondary prevention of cardiovascular disease in the United States. *Eur Heart J*. 2024;45(suppl 1):ehae666.

38. Lawson CA, Benson L, Squire I, et al. Changing health related quality of life and outcomes in heart failure by age, sex and subtype. *EClinicalMedicine*. 2023;64:102217.

39. Omidvari A-H, Lansdorp-Vogelaar I, de Koning HJ, Meester RG. Impact of assumptions on future costs, disutility and mortality in cost-effectiveness analysis; a model exploration. *PLoS One*. 2021;16(7):e0253893.

40. Boye KS, Thieu VT, Sapin H, et al. Patient-reported outcomes in people with type 2 diabetes receiving tirzepatide in the SURPASS clinical trial programme. *Diabetes Ther*. 2023;14(11):1833–1852.

41. Gudzune KA, Stefanski A, Cao D, et al. Association between weight reduction achieved with tirzepatide and quality of life in adults with obesity: results from the SURMOUNT-1 study. *Diabetes Obes Metab*. 2025;27(2):539–550.

42. National Academies of Sciences. Engineering, and Medicine, Health and Medicine Division, Board on Health Care Services, et al. In: *Making medicines affordable: a national imperative*. National Academies Press; 2017.

43. Hammond RA, Levine R. The economic impact of obesity in the United States. *Diabetes Metab Syndr Obes*. 2010;3:285–295.

KEY WORDS cardiometabolic, economic evaluation, health outcomes, Markov model, pharmacoeconomics

APPENDIX For an expanded Methods section as well as a supplemental figure and a table, please see the online version of this paper.